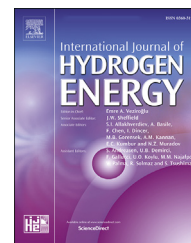


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Metal hydride reactor for output temperature control

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HIGHLIGHTS

- Precise temperature control required for extended lifetime of fuel cells.
- Metal hydride module suggested as active thermal energy buffer for T-control.
- 0D model identified and validated with experimental data of metal hydride reactor.
- Designed controller successfully demonstrates T-stabilization at 70 °C.
- Response time of fluid outlet temperature of MH reactor is < 5 s.

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ABSTRACT

Precise temperature control of fuel cell stacks is crucial to avoid degradation mechanisms and thus to extend lifetime. The present study demonstrates that metal hydride based modules can assist in controlling this temperature by serving as active thermal energy storage. The reversible reaction of metals with hydrogen is characterized by a temperature pressure correlation. Therefore, the temperature as well as the thermal power at which the thermal energy is released can be defined by the gas pressure. This enables the design of temperature controllers using the pressure as actuating variable on the metal hydride device. In this study, an existing metal hydride reactor based on $\text{LaNi}_{4.85}\text{Al}_{0.15}$ is integrated into a hydrogen and cooling fluid testing infrastructure. Based on experiments indicating the step response behaviour, the parameters for a 0D model have been identified. A PI controller was designed based on simulations and implemented on the test rig hardware. First experimental results show that it is possible to stabilize the temperature of the cooling fluid circuit to $70^\circ\text{C} \pm 1.5\text{ K}$ while a thermal power of $+755\text{ W/kg}_{\text{MH}}$ and $140\text{ W/kg}_{\text{MH}}$ is provided for absorption and desorption, respectively.

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1. Introduction

The commercialization of fuel cells has already started in the last decade, while there are still challenges for research like long-term durability of fuel cells in automotive applications [1]. Reducing degradation of fuel cells leads to a longer life time and thus to lower costs and higher sustainability of the fuel cell technology [2]. Fuel cells degrade in their performance over time due to mechanical, chemical and thermal mechanisms [3]. Thereby, mechanical failures have the highest occurrence potential and are considered as the main cause of early fuel cell failures [4,5]. In automotive applications the fuel cell is exposed to varying operation condition, leading to changes of the electrode potential, stoichiometric ratio of the supplied gases, internal temperature and humidity conditions [1]. Most of the mechanical degradation mechanisms as shrinking-swelling, fatigue, side chain scission processes and wrinkles are triggered by humidity fluctuations due to dynamic load changes or start ups and shutdowns [1,6]. For example low humidity and high temperatures lead among other aspects to an approximately ten time lesser lifetime of fuel cells in mobile application in comparison to operating as a fixed power source [7]. Furthermore, it could be shown that reducing the amplitude and frequency of humidity cycling leads to an increase of the membrane durability [8]. Therefore, in order to extend lifetime and reduce cost, the humidity and thus the temperature control of proton-exchange membrane fuel cell (PEM) is crucial and therefore extensively studied [9].

Deviations of the optimal temperature occur mainly during two characteristic events: frost/cold start during the start-up phase and load variations during operation [1]. To reduce the numbers of frost start/cold start, various techniques are studied [10], e.g., electrical heaters, increase of oxygen partial pressure, or metal hydride (MH) based preheaters [11]. The concept in this paper focus on the stabilization of the operation temperature during load variation.

There exist various cooling techniques for stabilization of the temperature during operation using air or liquid cooled systems, e.g., phase change or passive thermal elements [12]. A common layout for a liquid cooled temperature control is based on a bypass system with an air cooler in the cooling fluid circuit [13]. The thermal power of such a cooling system directly depends on the outside temperature as well as the surface of the cooler, which is an important factor for the packaging, e.g., in a vehicle application. In case of a temperature peak reduction in the cooling circuit, this cooler size could be reduced. So, the question arises, if there exists a system which could act as small thermal energy buffer providing a reduction of temperature peaks. Obviously, such kind of thermal energy buffer needs to fulfill requirements in size, weight and especially response time such that it could actually improve the existing simple cooling systems. In the present manuscript it will be shown that open MH systems are an option for such thermal buffers, as they show fast reaction kinetics and high thermal energy storage capacities. Furthermore, it is possible to integrate the system into the existing hydrogen infrastructure and using the gas pressure for active control, while no hydrogen is actually consumed [14]. The goal of this manuscript is to introduce the concept of such

an active MH temperature controller and demonstrate its performance in an experimental setup. The results provide first insights into the power output that can be achieved, the response time of the controller and pave the way for future work.

The operation principle underlying the suggested active thermal buffer is based on the reversible gas solid reaction of a MH with hydrogen (H) according to:



The absorption reaction of hydrogen is exothermal and the desorption reaction endothermal, expressed by the reaction enthalpy ΔH_R . The variable x and y stands for an arbitrary integer.

There exist plenty of studies in which MH tanks are coupled to fuel cells and the required energy balances are discussed ([15–18]) or the active control of the thermal management is considered ([19,20]). However, the majority of these studies refer to MHs as gas storage devices. Thus, the hydrogen discharging rates are rather slow in comparison to their capacity (e.g. complete discharging in 1 – 2 h).

The present MH device for active temperature control of the heat transfer fluid circuit is not considered as gas storage, but as a device that can provide high thermal powers for hydrogen based energy systems. In this case, the hydrogen gas is rather a requirement to release or provide heat and not to be stored itself.

Fig. 1 shows the intended integration of such an active thermal buffer as a so called open MH system [14]. As it can be seen, the module is integrated into the hydrogen supply line of the fuel cell, thus it can be assumed that at any time hydrogen can be supplied at a pressure of 8 bar to the system. Furthermore, if required, it can be released to a fuel cell at a few mbar above ambient pressure. This open metal hydride concept has already been presented for air-conditioning applications [21] and it relies on the fact that the energy effort for hydrogen compression is outside of the considered energy balance. As a result, the energy available in the compressed hydrogen is granted for free and in this system no hydrogen is consumed.

The operation principle of the intended temperature control by such a system integrated into an existing hydrogen infrastructure is also shown schematically in Fig. 1. Whenever the fuel cell is providing a varying electrical power demand, the thermal power released as well as the outlet temperature of the fuel cell vary accordingly (I). In order to compensate this temperature variation before it re-enters into the fuel cell, the metal hydride module is integrated into the heat transfer circuit. The pressure in the metal hydride module (II) can then be used to trigger the desired reaction and control the temperature in the MH (III): When the fluid temperature is below the set temperature, the pressure in the metal hydride reactor needs to be increased. Then, the exothermal reaction will take place and the temperature of the metal hydride increases. In case the fuel cell outlet temperature is above the set temperature, the hydrogen pressure will be decreased, resulting in the endothermal desorption and thus in a lower temperature. As a result, the outlet temperature of the MH reactor, which refers to the inlet temperature of the fuel cell, can be kept constant (IV).

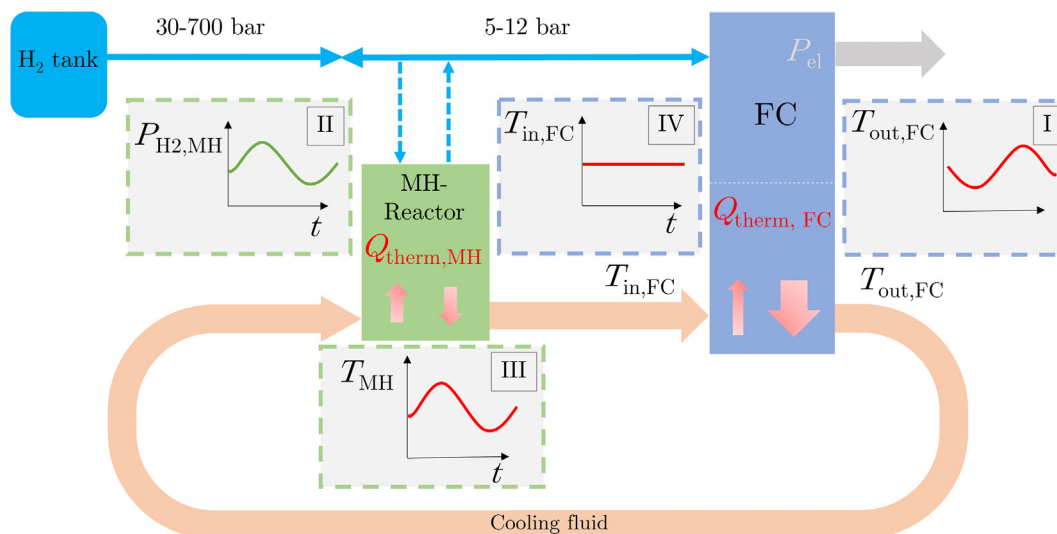


Fig. 1 – The figure shows the intended integration of an active thermal buffer (green) as a so called open MH system. The H_2 components are colored in bright blue, the fuel cell components in darker blue and the cooling circuit is colored in light orange. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article).

So overall, the envisioned concept is based on the following two characteristic features of the reaction. First, due to the reversible exo-/endothermal reaction when hydrogen is provided or removed, it is possible to provide or remove thermal energy from the system. Thus, at the outlet of the active thermal buffer element a constant fluid temperature can be achieved even though the inlet temperature is fluctuating. Second, the pressure-temperature relationship combined with the available gas pressure levels enables not only to control the temperature level at which heat and cold are produced, precisely. Rather, there is the option to release the heat at a much higher temperature than the cold is provided. Thus, the dynamics of this active thermal buffer for smoothing of fluctuating inlet temperatures can be adjusted as required.

In order to control the set pressure for an active metal hydride module, a corresponding active MH temperature controller can be described by a cascaded control structure

shown in Fig. 2. The outer control variable of the cascaded temperature controller is the outlet temperature T_{cf} of the cooling fluid (CF) at the end of the MH-reactor. This temperature is disturbed by a change in the inlet temperature $T_{cf,in}$, e.g., by the load change of the fuel cell. To compensate the disturbance and maintain a desired temperature, the outer control loop calculates a desired pressure p_{set} from the error e_T . A pressure controller in the inner control loop controls the measured hydrogen pressure p_g far from the inlet of the MH reactor by adapting the pressure p_R at the inlet of the reactor. The error e_p stands for the difference between the pressure p_g and p_{set} . The mass flow $\dot{m}_{cf}$ of the cooling fluid is assumed to be constant in this article. Using the gas pressure as actuating variable, the temperature of the MH is set imposing two effects on the apparent thermal power. First, the absolute value of the temperature in the MH affects the reaction rate and thus the amount of heat that is released or absorbed. Second,

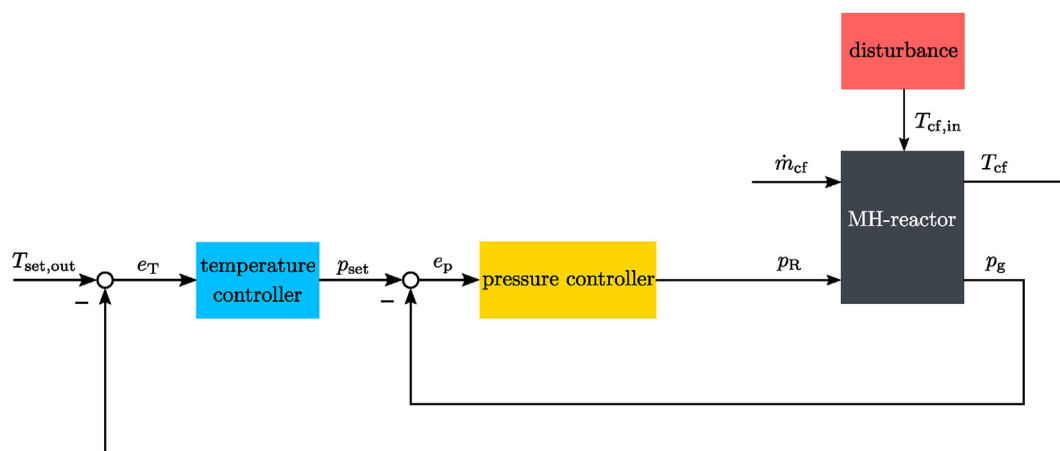


Fig. 2 – Scheme for describing a cascaded active MH temperature controller.

the emerging temperature gradient between the MH and the heat transfer fluid determines the driving force for the heat transfer. The research questions are: which temperature gradients can be compensated, at which specific thermal power and what is the response time of the temperature signal following the gas pressure?

In order to demonstrate the proof of concept, a reactor designed for high thermal powers is required. In the present study, an existing reactor reported in Ref. [22] for preheating of fuel cells has been used. The reactor is filled with 960 g of the MH $\text{LaNi}_{4.85}\text{Al}_{0.15}$, which corresponds to an estimated thermal energy of 192 kJ for a full absorption reaction and 209 kJ for a full desorption reaction. The material is suitable for the study, as its equilibrium pressure at 70 °C and 50% hydrogen loading is 5.7 bar. Assuming that in a fuel cell infrastructure hydrogen is available at 8 bar while it is required in the fuel cell at few mbar above ambient pressure, the MH can achieve temperatures between 20 °C and 75 °C with a hydrogen loading of 90%. Details on the material properties especially the conversion dependence of the equilibrium conditions can be found in Ref. [23].

2. Experimental testbench and model description

2.1. Experimental set-up

To demonstrate the performance of the temperature controller, an existing test bench with a MH reactor was

adapted for identification and control of the reactor from Ref. [24]. The setup of the test bench can be found in Fig. 3.

It consists of the cooling fluid circuit in black and the hydrogen gas part in blue. The cooling fluid flows from the thermostatic bath through the pump, the needle valve NV_1 and through the reactor back to the bath. Thus, in this setup, the thermostatic bath represents the fuel cell in a later application, which can provide varying outlet temperatures at various electrical loads. The two thermocouples (index small c, T_c) measure the cooling fluid temperature at the inlet $T_{c,cf,in}$ and the outlet $T_{c,cf}$ of the reactor. NV_1 regulates the cooling fluid mass flow. The regulated mass flow of the cooling fluid is measured by a Coriolis mass flow meter $\dot{M}_{H_2O}$.

The MH reactor connects the cooling fluid circuit with the hydrogen circuit. The reactor is based on a shell-and-tube heat exchanger and was designed and build in Ref. [24]. It has one inlet and one outlet for the cooling fluid (black) and two ports on the hydrogen side (blue). At one hydrogen port, the pressure sensor P_4 with the measured pressure p_r is attached. The other port serves as in- and outlet for the hydrogen to the reactor. Therefore, the port is connected with the blue hydrogen circuit.

The hydrogen for the experiments is provided from a hydrogen reservoir and is buffered in the Sieverts volume for precise balancing. To allow the balancing, the pressure sensor P_1 measures the pressure p_{siev} and the thermocouple $T_{c,siev}$ the temperature T_{siev} in the Sieverts volume. The Sieverts volume is connected to the inlet and outlet of the reactor by the pressure regulator PR_1 . Further the inlet and outlet of the

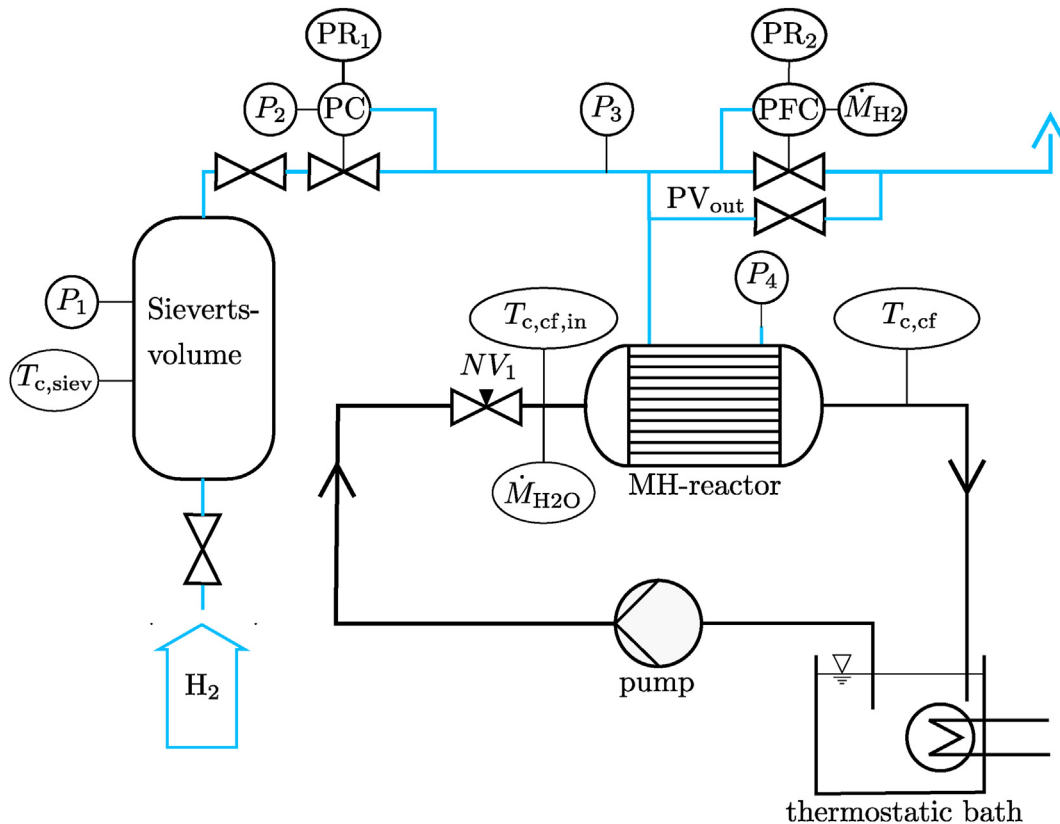


Fig. 3 – Illustration of the test bench setup with the cooling fluid circuit in black and the hydrogen part in blue. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article).

reactor is connected with the pressure regulator and flow meter PR_2 . PR_2 controls the pressure for an outflow of hydrogen from the reactor to the surrounding. The outflowing mass flow is measured by the mass flow meter $\dot{M}_{H_2}$. The maximum measurable flow of the sensor is limited to 5 Nl/min. Therefore, a bypass valve PV_{out} for higher not measured outflows is included. The pressure sensor P_3 measures the pressure in the pipes p_p . Table A.3 of the appendix gives an overview on the measurement equipment.

Using the measured values for temperature, pressure and mass flows the heat flow from the MH to the cooling fluid, the mean temperature and the hydrogen charging are calculated. The hydrogen charging ω is calculated differently for the two types of reactions. This is due to the different measurement methods during absorption and desorption.

During absorption, ω is calculated by integrating the changes in the amount of hydrogen in the Sieverts volume between time steps. During desorption the hydrogen flow can be measured with PR_2 for hydrogen flow rates below 5 Nl/min and recalculated with the ideal gas law to a hydrogen charging. For higher volume flows, the hydrogen charging is estimated through the pressure. Therefore, the hydrogen charging value at the beginning and the end of the reaction is determined and a relation between pressure and the hydrogen charging is adopted. The exact calculation of the hydrogen charging during absorption and desorption can be comprehended in the Chapter Appendix B.

Finally, the heat flow between the MH and the cooling fluid from measurement (meas) data can be calculated as

$$\dot{Q}_{cf,meas} = c_{p,cf} \dot{m}_{cf} (T_{cf} - T_{cf,in}) - \dot{Q}_{loss}, \quad (2)$$

with the heat loss to ambience $\dot{Q}_{loss}$, the specific heat capacity of the cooling fluid $c_{p,cf}$ and the mass flow of the cooling fluid $\dot{m}_{cf,meas}$. The heat loss refers to a constant value for the considered temperature range $T_{cf,in} = [60, 70, 80]^\circ\text{C}$. It is determined during steady state at $T_{cf,in} = 70^\circ\text{C}$ with an inactive reactor ($\dot{Q}_{cf,meas} = 0$). The heat loss has to be considered in the experimental data, as the model of the reactor will be discussed without the influence of the ambient.

2.2. Model description

For the mathematical modeling approach, the reactor described in the previous section is simplified as a lumped 0D-model with only one tube. Further, the reactor is divided into three different balance areas: hydrogen gas (light blue), cooling fluid (dark blue) and the hollow cylinder of the MH (purple). The lumped 0D-model is considered without influence of the ambient. The balance areas and the physical quantities for setting up the model equations are shown in Fig. 4.

The balance for the cooling fluid area is derived from the energy balance described with the density of the cooling fluid ρ_{cf} , the volume of the cooling fluid inside the reactor V_{cf} and the time derivative of the temperature T_{cf} as

$$\rho_{cf} c_{p,cf} V_{cf} \frac{\partial T_{cf}}{\partial t} = \dot{Q}_{in} - \dot{Q}_{out} - \dot{Q}_{MH,cf}. \quad (3)$$

The inlet heat flow $\dot{Q}_{in}$ with the inlet temperature $T_{cf,in}$

$$\dot{Q}_{in} = c_{p,cf} \dot{m}_{cf} T_{cf,in} \quad (4)$$

and the outlet heat flow $\dot{Q}_{out}$ with the outlet temperature T_{cf}

$$\dot{Q}_{out} = c_{p,cf} \dot{m}_{cf} T_{cf} \quad (5)$$

are calculated with the mass flow rate $\dot{m}_{cf}$ and the constant heat capacity of the cooling fluid $c_{p,cf}$. Further, the heat flow $\dot{Q}_{MH,cf}$ from the metal hydride to the cooling fluid is calculated by

$$\dot{Q}_{MH,cf} = k A \Delta T_{MH,cf}, \quad (6)$$

with the heat transfer coefficient k , the heat transfer surface A and the driving temperature difference $\Delta T_{MH,cf}$. The driving temperature difference is calculated using the assumption of a logarithmic temperature difference, as the representative fluid temperature is changing during the reaction [25].

$$\Delta T_{MH,cf} = \frac{T_{MH} - T_{cf} - (T_{MH} - T_{cf,in})}{\ln \left| \frac{T_{MH} - T_{cf}}{T_{MH} - T_{cf,in}} \right|}, \quad (7)$$

with the temperature of the MH T_{MH} .

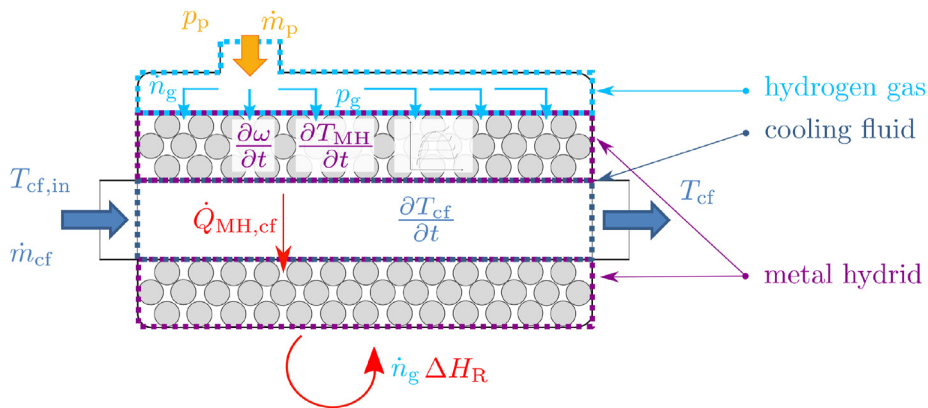


Fig. 4 – Illustration of the reactor with the defined balance areas. Based on [24].

For setting up the MH balance as energy balance, the heat flow $\dot{Q}_{MH,cf}$ and the reaction enthalpy flow $\dot{n}_g \Delta H_R$ are considered (see Fig. 4). The reaction enthalpy flow consists of the consumed amount of hydrogen $\dot{n}_g$ and the reaction enthalpy ΔH_R . With those heat flows the equation for the respective balance area can be described by the time derivative of the temperature in the MH T_{MH} , the mass of the MH m_{MH} and the specific heat capacity $c_{v,MH}$

$$m_{MH} c_{v,MH} \frac{\partial T_{MH}}{\partial t} = \dot{Q}_{MH,cf} + \dot{n}_g \Delta H_R. \quad (8)$$

To describe the dynamic of the MH reaction by a pressure or temperature change, a kinetic equation has to be introduced. A simple approach for gas-solid-reactions as in Ref. [23] is

$$\frac{\partial \omega}{\partial t} = \underbrace{K_0 e^{\frac{-E_a}{RT_{MH}}}}_{f(T)} \underbrace{\left(\frac{p_g - p_{eq}}{p_{eq}} \right)}_{f(p)} f(\omega) \quad (9)$$

where $\partial \omega / \partial t$ describes the hydrogen charging rate through a temperature, pressure and charging dependent part. The temperature dependent part $f(T)$ is based on an Arrhenius approach with the activation energy E_a , the ideal gas constant R , the temperature of the MH bed T_{MH} and the pre-exponential factor K_0 . In this case, the pre-exponential factor represents the Arrhenius factor K_{Ar} of the material kinetics and as well unmodeled influences, namely heat and mass transfer effects in the lumped system. The pressure dependent part $f(p)$ describes the normalized driving force by the pressure difference in the MH bed p_g and the equilibrium pressure p_{eq} . This difference is normalized with the equilibrium pressure according to Ref. [26]. The reaction mechanism $f(\omega)$ to describe the influence of the hydrogen charging, distinguishes between absorption and desorption as

$$f(\omega) = \begin{cases} \omega_{max} - \omega & \text{absorption} \\ \omega & \text{desorption.} \end{cases} \quad (10)$$

In the following section a description for the parameters $\dot{n}_g$, p_g and p_{eq} is derived.

With the derived Eq. (9) for the dynamic of the hydrogen charging ω , the molar flow rate $\dot{n}_g$ can be described by

$$\dot{n}_g = \frac{\partial \omega}{\partial t} \frac{m_{MH}}{100 M_{H_2}}. \quad (11)$$

For describing the pressure of the hydrogen gas p_g in Eq. (9), a mass balance considering the mass flow rate into the reactor $\dot{m}_p$ and the reaction hydrogen molar flow rate $\dot{n}_g$ (replaced by (11)) is set up. With the assumptions of ideal gas behaviour, constant volume of the radial gas gap V_g and constant ambient temperature T_{amb} of the hydrogen gas, the pressure in the balance area can be described as

$$\frac{\partial p_g}{\partial t} = \frac{R T_{amb}}{M_{H_2} V_g} \left[\dot{m}_p - \frac{\partial \omega}{\partial t} \frac{m_{MH}}{100 M_{H_2}} \right]. \quad (12)$$

The mass flow rate $\dot{m}_p$ into the reactor and the void volume of the tubes is approximated with a polytropic and reversible adiabatic outflow equation of an ideal gas at T_{amb} , as it is assumed that this temperature is representative for the incoming gas. The derivation can be found in Appendix C.

For MH (based) reactions, temperature, pressure and charging state are closely correlated, thus a mathematical description of this relationship is required as baseline for the mathematical model. Using enthalpy and entropy data from Ref. [24], for the present material, the following description has been deduced

$$\ln \left(\frac{p_{eq}}{p_0} \right) = \begin{cases} \frac{\Delta H_{abs}(\omega)}{R T_{MH}} + \frac{\Delta S_{abs}(\omega)}{R} + \ln(I_{add}(\omega)) & \text{Absorption} \\ \frac{\Delta H_{des}(\omega)}{R T_{MH}} + \frac{\Delta S_{des}(\omega)}{R} & \text{Desorption} \end{cases} \quad (13)$$

which distinguishes between absorption (abs) and desorption (des) with an additional proportion $I_{add}(\omega)$ for the steep increases of the MH plateaus. In the equation, T_{MH} represents the temperature of the MH, p_0 the standardized pressure of 1 bar, R the ideal gas constant, ΔH_R and ΔS_R the reaction enthalpy and entropy. Eq. (13) is valid for $0 < \omega < \omega_{max}$ and $p_0 > 0$. For more details refer to Appendix D.

Table 1 lists all parameters of the balance equations.

Table 1 – Overview of the symbols, values and references for the model parameters and constants. When there is no reference given, the value is measured or a general known constant.

Parameter	Symbol	Value	Reference
Parameters MH			
MH mass	m_{MH}	0.96 kg	[24]
Specific heat capacity MH	$c_{v,MH}$	419 J kg ⁻¹ K ⁻¹	[27]
Parameters cooling fluid water			
Specific heat capacity water	$c_{p,cf}$	4189, 6 J kg ⁻¹ K ⁻¹ at 70 °C	[28]
Density water	ρ_{cf}	977, 76 kg/m ³ at 70 °C	[28]
Parameters reactor			
Cross sectional area of hydrogen inlet	A_g	25 mm ²	
Volume hydrogen reactor	V_g	191 cm ³	
Volume cooling Fluid in reactor	V_{cf}	1.4523 × 10 ⁻⁴ m ³	[24]
Constants			
Ideal gas constant	R	8, 314 J mol ⁻¹ K ⁻¹	
Molare mass hydrogen	M_{H_2}	2, 016 g mol ⁻¹	

3. Results and discussion

In the following subsection, the parameters of the model Eqs. (3), (8) and (9) are identified and the model is validated with experimental data. The main focus in the identification is on the dynamic between the hydrogen pressure and the MH hydrogen loading values. In the subsequent section, the model with the identified parameters will be used for the design of the controller.

3.1. Problem set-up of the parameter identification

In the first part of this section, the parameter vector Θ

$$\begin{aligned}\Theta &= [\Theta_1 \quad \Theta_2 \quad \Theta_3 \quad \Theta_4 \quad \Theta_5] \\ &= [kA \quad E_a \quad K_0 \quad k_{\text{peq}} \quad \omega_{\text{max}}],\end{aligned}\quad (14)$$

which is to be identified will be explained using the model equations.

The first parameter for identification corresponds to the heat flow from the cooling fluid to the MH. In Eq. (6) the temperature gradient, the surface A and the heat transfer coefficient k describe this one dimensional simplified heat flow from the cooling fluid to the MH. For the present reactor design an analytical estimation of A and k would be imprecise due to missing parameters such as contact resistance or local power density. That is why the product of $k \cdot A$ is identified as lumped parameter from experimental data.

Regarding the reaction rate in Eq. (9), the activation Energy E_a and the lumped parameter K_0 are identified as well. Thereby, E_a represents the physical property of the energy barrier, which is necessary to start the reaction. K_0 presents the conventional Arrhenius factor and includes unmodeled effects of the 0D model in mass and heat transfer.

The next two parameters for the identification are model correction parameters as the 0D model of the present reactor is simplified. The model depends on material properties, which are functions of pressure and temperature of the MH. In this work, it is assumed that the inlet temperature $T_{\text{cf,in}}$ corresponds to an overall average of the MH temperature in the reactor. This assumption leads to errors in the corresponding calculated material properties, as the actual temperature of the MH will be actually lower than the assumed temperature. As a result also the actual equilibrium pressure will be lower than the calculated one. This effect could be seen in simplified pressure concentration isotherm (PCI) measurements during absorption. In order to correct this effect in the model, the fitting parameter k_{peq} is introduced during absorption reactions.

In the evaluation of the measurement data, it can be observed that the maximum hydrogen charging state ω_{max} is smaller than expected from literature data of [24]. To incorporate the observed lower maximum values, the parameter ω_{max} is identified.

Next to the model parameters, the excitation signals have to be chosen for the parameter identification. The reaction of MHs with hydrogen generally represents a stable system with an equilibrium between gas pressure, temperature and hydrogen charging. This equilibrium of MH can be changed by

forcing a temperature or pressure change from outside, which will lead to a (partial) reaction and thus to a defined fraction of the absorption or desorption reaction enthalpy. Since in the present system gas transport is much faster than heat transfer, the pressure p_g is chosen as excitation signal.

In the next step, the type of excitation signal needs to be determined. To keep the identification simple, pressure step functions with different starting and final values, which are based on fixed hydrogen loading values, are used in the present study. Further, for a temperature control it is important to know how much cold or heat can still be released by the MH, why the hydrogen loading state and not the heat flow is used for identification.

As a consequence, for the experimental design of the measurements the following procedure has been defined. The baseline experiments take place at 60 °C and 80 °C. Experiments at 70 °C are used for validation. For each temperature four different pressure levels for absorption and desorption reaction are chosen. An overview over the resulting pressure profile is given in Fig. 5, where the absorption reaction with the pressure steps leading to changes in the hydrogen loading from 0% to 10%, 10%–50%, 50%–90%. The start and endpoint pressures of the step functions are determined from the measured data from Ref. [24]. The hydrogen loading value of 0% was set at a pressure of 1 bar. The step from 0% to 100% is called “full step”. The expression “partial step” is used for all other changes in the hydrogen loading, e.g., 10%–50%.

3.2. Parameter identification

As mentioned before, the control objective for the current system is the stabilization of the reactor output temperature from the heat transfer fluid to a given set point. Therefore, a cost functional for the following parameter identification of the reactor model is defined. The cost functional takes two aspects into account: The hydrogen charging state ω and the thermal energy transferred from/to the heat transfer fluid $\dot{Q}_{\text{MH,cf}}$. The transferred heat directly influences the outlet temperature of the reactor. The hydrogen charging state has a

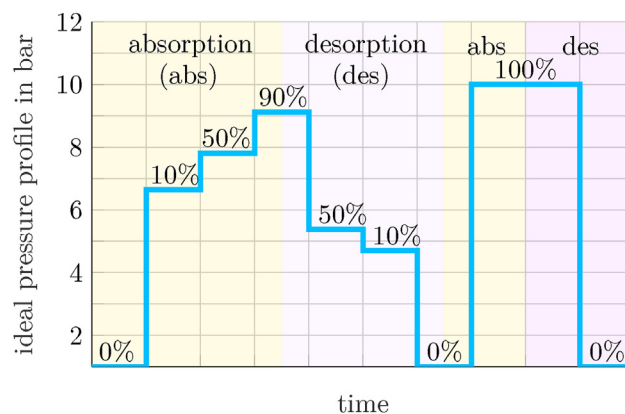


Fig. 5 – Exemplary representation of the pressure steps in a row for the temperature 80 °C. The aimed hydrogen loading of a pressure jump is written at the end of each pressure step.

strong influence on the dynamics of the system and additionally determines the amount of hydrogen that can be absorbed or desorbed starting from the current state. Moreover, reaction entropy and the kinetics are also dependent on the actual hydrogen charging. For the optimization problem including all mentioned equations the following state vector

$$\mathbf{x} = [T_{cf} \quad \omega \quad T_{MH}] \quad (15)$$

and input vector is defined as

$$\mathbf{u} = [p_g \quad T_{cf,in}]. \quad (16)$$

With those vectors the least square formulation of the cost functional is set up as the following optimization problem

$$\min_{\Theta} \sum_{t_k=0}^{t_N} 0.4 \frac{(\omega_{sim}(t_k) - \omega_{meas}(t_k))^2}{\omega_{norm}} + 0.6 \frac{(\dot{Q}_{cf,sim}(t_k) - \dot{Q}_{cf,meas}(t_k))^2}{\dot{Q}_{norm}} \quad (17)$$

s.t.

$$0 \leq x_2 \leq \Theta_5$$

$$\mathbf{x}(0) = [x_{1,0}, x_{2,0}, x_{3,0}]$$

$$\dot{x}_1 = \frac{1}{\rho_{cf} c_{p,cf} V_{cf}} [-\Theta_1 \Delta T_{MH,cf} + \dot{m}_{cf} c_{p,cf} (u_2 - x_1)]$$

$$\dot{x}_2 = \Theta_3 e^{\frac{-\Theta_2}{R x_3}} \left(\frac{u_1 - p_{eq}}{p_{eq}} \right) f(x_2)$$

$$\dot{x}_3 = \frac{1}{m_{MH} c_{V,MH}} \left[\Theta_1 \Delta T_{MH,cf} + \dot{x}_2 \Delta H_R \frac{m_{MH}}{100 M_{H_2}} \right]$$

$$\Delta T_{MH,cf} = \frac{x_3 - x_1 - (x_3 - u_2)}{\ln \left| \frac{x_3 - x_1}{x_3 - u_2} \right|}$$

absorption

$$p_{eq} = p_0 e^{\frac{\Delta H_{abs}(x_2)}{RT} + \frac{\Delta S_{abs}(x_2)}{R}} + I_{add} - \Theta_4$$

$$\Delta H_R = \Delta H_{abs}$$

$$f(x_2) = (\Theta_5 - x_2)$$

desorption

$$p_{eq} = p_0 e^{\frac{\Delta H_{des}(x_2)}{RT} + \frac{\Delta S_{des}(x_2)}{R}}$$

$$\Delta H_R = \Delta H_{des}$$

$$f(x_2) = x_2$$

in between

$$f(x_2) = 0.$$

(17)

The cost functional in the optimization problem evaluates the difference between the measurement data (ω_{meas} , $\dot{Q}_{cf,meas}$) and the simulation data (ω_{sim} , $\dot{Q}_{cf,sim}$) at the discrete time points t_k . To avoid weighting due to different magnitudes the heat flow and the hydrogen charging are normalized accordingly by ω_{norm} and $\dot{Q}_{norm}$. Both values represent the maximum value during the first step of the optimization. The error in the heat flow is weighted higher in the cost functional as the error in the hydrogen charging. The weighting is a result of parameter studies with selected different weights. The cost functional is restricted by the model Eq. (3) for $\dot{x}_1$, (9) for $\dot{x}_2$, (8)

for $\dot{x}_3$, (7) for $\Delta T_{MH,cf}$, (13) for p_{eq} , (10) for $f(\omega)$ and the calculation of the heat stream from measurement data in Eq. (2). A distinction of cases is performed for the equilibrium pressure, the reaction enthalpy and the term $f(x_2)$. Further, the calculation of the equilibrium pressure of the MH in Eq. (13) is supplemented by the identification parameter Θ_4 .

The optimization problem with the given input and state vector is solved with the pattern search algorithm `PATTERN-SEARCH` in `MATLAB/SIMULINK`, while the model is simulated with the solver `ODE23s`.

The optimal result for the parameter vector Θ is given by

$$\Theta^* = [kA^* \quad E_a^* \quad K_0^* \quad k_{peq}^* \quad \omega_{max}^*] \quad (18)$$

$$= [735.5 \quad 36000 \quad 12016 \quad 48448 \quad 1.175]$$

using the measurement data for 60 °C and 80 °C. This temperatures are chosen for identification due to the assumption that most of the fuel cells will run between this temperatures. Comparing the result to the literature, the optimal parameter values are in physically meaningful ranges e.g., ω_{max} can be estimated from Ref. [24] with approximately 1.3 wt-% or E_a with approximately 36 202 J/mol. The optimization achieves a coefficient of determination R^2 of 0.88 for ω and 0.81 for $\dot{Q}$.

3.3. Model validation

As the model parameters have been identified using the measurement data at 60 °C and 80 °C, for validation, the simulation results of the identified model parameters at 70 °C are compared to the measurement data. In Fig. 6, a comparison of the reaction rates is presented. The model represents the same trends like an increase in reaction rate with a lower initial charging state as can be seen in the measurement values. Furthermore, the general shape of the curves can be compared between measurement and simulation. However, overall the simulations are delayed and have a damped amplitude compared to the measurements.

This deviation in the modeling of the reaction rate consequently also causes deviations in the determined charging state. In Fig. 7, from the start up to minute 92, three partial step absorption reactions take place which can be observed from the measured as well as from the simulated data. During those reaction periods, a mean error between the measurement and simulation data of 10.4% occurs. As mentioned before, this error is mainly due to the deviations in the reaction rate. Further, the measured pressure p_r is used for the simulation. The pressure sensor P_4 , which measures this pressure p_r , is positioned far from the hydrogen inlet in the reactor. That's why the measured pressure p_r changes more slowly than the mean reactor pressure, resulting in a reaction delay in the beginning.

Starting from 92 min, three desorption reactions are visible. In comparison to the absorption reactions, a similar error of 11.4% can be observed. The differences are mainly due to the necessary estimation (limited outflow of PR_2) of the hydrogen conversion rate during desorption and a lack for PCI data in the examined temperature range.

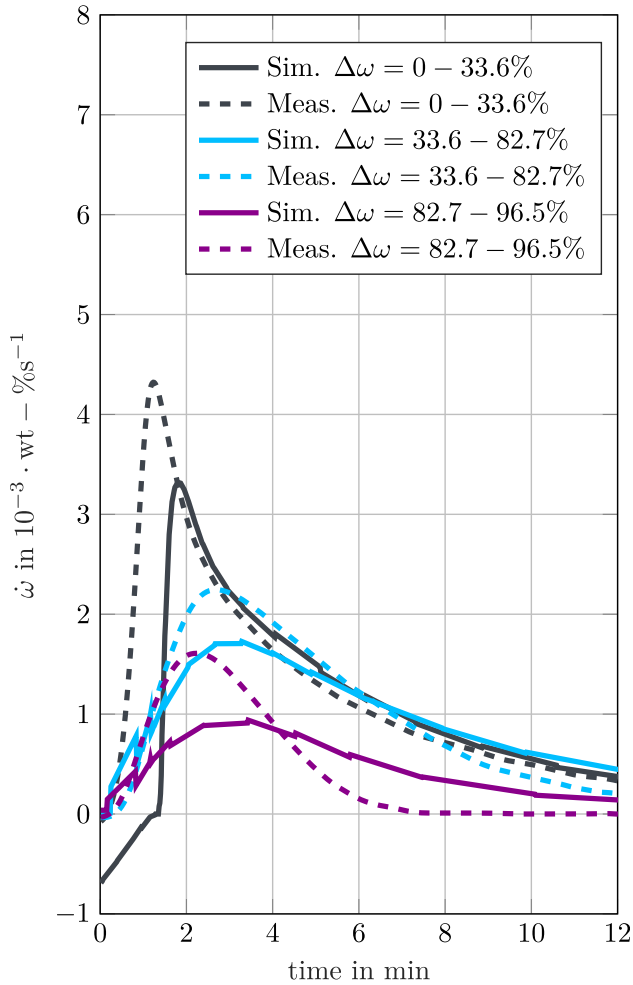


Fig. 6 – Comparison between simulation and experimental data of the reaction rate for different partial pressure steps.

In comparison to the reaction rates, the heat flow determined for measurement and simulation fits quite well. The mean error during the partial step absorption in Fig. 7 is about 3.9% and the mean error during the full step desorption 9.2%. The peaks in the error at the beginning of the reactions are again due to a slower start of the reaction in simulation. A delayed change of the hydrogen charging corresponds to a delayed start of the reaction (due to the pressure assumption) and thus to a delayed heat flux.

In conclusion, the model captures the general underlying trends that are required for the development of the control strategy in the next section. However, there is still a difference between the maximum hydrogen state $\omega_{\max}$ and the identified one. In order to further improve the model quality regarding hydrogen charging rate and charging state, improved data for PCI would be required as well as more representative measurements probes for lumped temperature and pressure values inside the reactor.

3.4. Temperature control

The main goal of the present publication is the demonstration of a control strategy for a metal hydride reactor in order to develop an active thermal buffer module. In this section, the

validated model will be used to develop a PI controller that is validated by additional experiments.

3.4.1. Controller design

The introduced cascaded control structure in Fig. 2 has the goal to stabilize the output temperature of the reactor due to temperature fluctuations of the cooling fluid and constant mass stream through the reactor. To achieve this goal, reaction enthalpy of the metal hydride must be released or removed from the fluid depending on the required heat source or heat sink to compensate the fluctuating inlet temperature signal, respectively. Obviously, the amount of enthalpy released or provided can be determined by comparison of the hydrogen charging state at the start and end of the corresponding fluctuating signal. To measure and define these changes of the hydrogen charging state during a reaction, volume flow sensors and pressure sensors combined with material data would be a simple solution. However, for application purpose the measurement of the volume flow of hydrogen is not practicable. Instead, more simple pressure sensors P_2 , P_3 and P_4 are used in the present test bench and the hydrogen charging is approximated over the pressure. The second inner control loop controls the pressure in the test bench. It consists of the chosen input variable p_{set} and p_g as control variable. The cascaded control structure has the advantages of achieving higher accuracy of the much faster varying pressure in the inner loop. That helps to reduce errors in the outlet temperature faster.

As in the test bench two pressure regulators are installed, the typically cascaded control structure from Fig. 2 is extended to the structure in Fig. 8. The location of the pressure regulators PR_1 and PR_2 can be seen in Fig. 3. The regulator PR_1 is integrated between the Sieverts volume and the reactor, the second regulator PR_2 is integrated between the reactor and the outlet to the surrounding. The parameters of the inlet regulator had to be identified, as they have not been provided by the device itself. As an approach a PI controller is used and the parameters of the controller are identified with pressure steps from 1 bar to 6 bar and 8 bar or 10 bar. The parameters of the outlet pressure regulator are determined from the programmable logic controller (PLC) software and the parameter sets for both pressure regulators are shown in Table 2. Depending on the difference between the given pressure p_{set} and the pressure inside the reactor p_g the PLC at the test bench switches between those two regulators.

Now the controller for the outer control loop for inlet temperature stabilization is designed. Therefore, the pressure p_g is added to the state vector (15) with the state Eq. (12) and the new input pressure p_R is added in the input vector (16). Thus, the new state vector can be introduced.

$$\mathbf{x}^* = [T_{\text{cf}} \quad \omega \quad T_{\text{MH}} \quad p_g] \quad (19)$$

and input vector

$$\mathbf{u}^* = [p_R \quad T_{\text{cf}}, \text{in}]. \quad (20)$$

As controller a PI control structure is implemented in MATLAB/SIMULINK embedded in the simulation of the whole control structure in Fig. 8. With the simulation model, the parameters K_p and K_i of the outer PI controller are tuned by the minimization of the cost functional

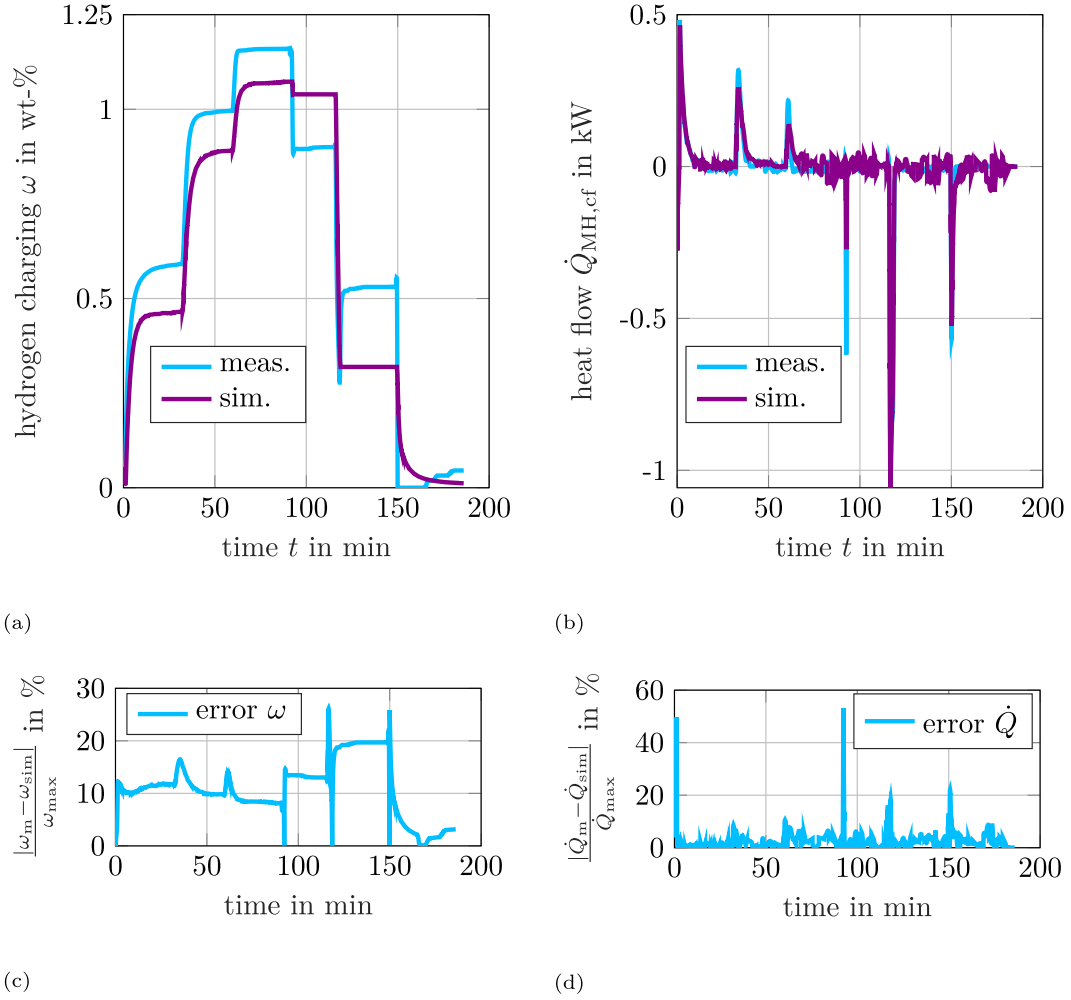


Fig. 7 – Validation of different small pressure steps in the simulation with measurement data at 70 °C. (a) H₂ charging for partial step response. (b) Heat flow for partial step response. (c) Relative error between the simulation and measurement data for the hydrogen charging. (d) Relative error between the simulation and measurement data for the heat flow.

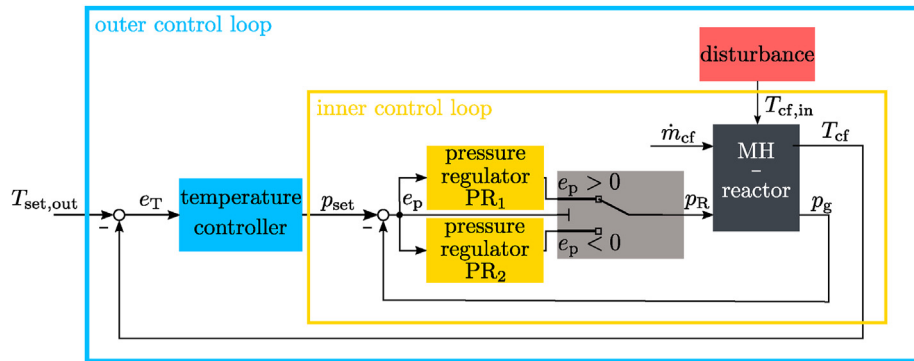


Fig. 8 – Extended control structure for describing an active MH temperature controller.

$$\begin{aligned}
 &\min_{K_p, K_i} \sum_{t_k=0}^{t_N} 0.4 \frac{(x_1(t_k) - T_{set,out}(t_k))^2}{e_{quad,norm}} + 0.6 \sum_{t_k=0}^{t_N-1} \frac{\dot{x}_1(t_k)}{e_{diff,norm}} \\
 &\text{s.t.} \\
 &\mathbf{x}^*(0) = [x_{1,0}^*, x_{2,0}^*, x_{3,0}^*, x_{4,0}^*] \\
 &\dot{\mathbf{x}}^* = f(\mathbf{x}^*, \mathbf{u}^*, K_p, K_i).
 \end{aligned} \tag{21}$$

The cost functional is inspired by Ref. [29] and consists of two weighted terms. The first term corresponds to the quadratic error of the difference between the desired set temperature $T_{set,out}$ and the temperature T_{cf} at the outlet of the reactor. The term is normalized with the difference of those two variables $e_{quad,norm}$ at the initial solution of the optimization. The quadratic error term of the temperature

Table 2 – Parameters of the inlet and outlet pressure controller.

Parameter	PR ₁	PR ₂
K _p	0.27	10
T _i	0.3	4
T _d	–	1
T ₁	–	0.2
Sampling time	0.1	0.1

leads to less deflection and a fast transient response of the control variable T_{cf} . The second term in the cost functional includes the time derivative of the quadratic error T_{cf} , which leads to a more damped behaviour. The term is normalized with $e_{diff, norm}$, representing the maximum value of $\dot{x}_1$ at the initial solution of the optimization. Both terms are evaluated at the discrete time steps t_k . The weighting factors are a result of a parameter study, where the weighting factors with the lowest deviation between the desired set temperature $T_{set,out}$ and the temperature T_{cf} at the outlet of the reactor have been selected. As constraints the model equations $f(\mathbf{x}^*, \mathbf{u}^*, K_p, K_i)$ are describing the reactor model with Eqs. ((3), (7)–(10), (12), (13), the PI-controller equations from PR₁, PR₂ and the gray colored switching element in Fig. 8. Using the cost functional introduced before, a disturbance is necessary to calculate a set of parameters.

The disturbance in the inlet temperature is adapted to the boundary conditions of the MH during absorption (maximum hydrogen charging, maximum heat flow) and the boundary conditions of the test bench (maximum measurable outflow). During desorption only a small temperature variance can be handled because of the limited hydrogen outflow of the pressure controller at the outlet. Thus, a trapezoidal signal, shown in Fig. 9(a), with a temperature decrease of 2.5 K for absorption and a temperature increase of 1 K for desorption is applied as a disturbance. A hold time of 30 s is provided when the target temperature is reached. Note, that the trapezoidal signal has the same time length for the temperature increase and decrease, what leads to different amounts of released thermal energy between desorption and absorption.

As a result the control parameters $K_p = 4.11$ and $K_i = 1$, 55 are determined for a sampling time of the temperature controller of 0.5 s. Using these parameters on the test bench, another experiment for the controller validation has been performed where the identical disturbance has been applied as in the simulations for the design of the controller.

3.4.2. Validation of the controller

Fig. 9 shows the results of the comparison between simulation (dark blue) and experimental results (light blue). The cooling fluid inlet temperature T_{cf} (yellow) shows the disturbance that has been imposed on the system around the set point temperature of 70 °C (black). This signal can be used to split the following discussion in two phases. First of all, from $t \approx 2$ min to $t = 10$ min, there is the negative disturbance in T_{cf} , which in turn requires the exothermal absorption process to stabilize the temperature. Second, from $t = 10$ min to the end of the experiment, a positive disturbance signal is applied to the

system which triggers the endothermal desorption reaction. For both time frames, it can be observed that the simulated temperature signal (light blue) as well as the measured cooling fluid temperature (dark blue) shows a temperature stabilization with minor oscillations around the set point of 70 °C. Thus, the implemented controller and the metal hydride reactor are able to proof the basic concept of the active thermal buffer.

The oscillations in simulation and experimental temperature profiles can be traced back to the two pressure controllers with their different dynamical behavior. Further, the oscillations during the absorption reactions are larger due to the larger inlet temperature disturbance and because of the higher possible dynamics of the mainly active pressure regulator PR₁. Due to inaccuracies in the modeling and the model based parameter tuning of the controller, the temperature controller is acting too aggressively in the measurement in comparison to the simulation.

Fig. 9(b) shows the pressure setpoint for the temperature controller in the simulation and the measurement. First of all, in this diagram, the oscillations in the measurement especially during the absorption phase have a large amplitude. The reason for these oscillations are the differences in the modeled and experimental reaction rate, leading to underestimated reaction rates from the temperature controller perspective. Due to these point, the designed parameters of the temperature controller are acting too aggressive in the experiments (see before). However, in the following the general trends in the set pressure will be discussed neglecting the oscillating behaviour.

During absorption phase, the setpoint of the pressure varies, although the set temperature is constantly at 70 °C. This indicates two important characteristics of the MH - hydrogen reaction system. First, while the inlet temperature continues to decrease to its minimum (until $t = 5$ min), the pressure increases. This increase in set pressure leads to a higher equilibrium temperature inside of the material and consequently to higher temperature gradient between the material and the cooling fluid. Thus, the heat transfer can be increased by this set pressure in order to stabilize the temperature to 70 °C despite the fluctuating inlet temperature. As mentioned in the introduction, this behaviour is quite special for thermal energy storages and can only be realized with thermochemical thermal energy storages as they show this pressure/temperature relation, which enables the design of such an active thermal energy buffer. Second, it can be observed that although the disturbance signal is symmetric, the set pressure at e.g. 3 and 8 min shows different values for the identical thermal power that needs to be transferred to stabilize the temperature. This behaviour is due to the fact that the pressure-temperature relationship is also a function of hydrogen charging and the hydrogen charging also varies during the process, which can be seen in Fig. 9(c).

Overall, it can be stated that the model is capturing the crucial dynamics of the MH reactor in a sufficient way that the resulting controller is able to realize the desired temperature stabilization. The strong oscillations that can be observed

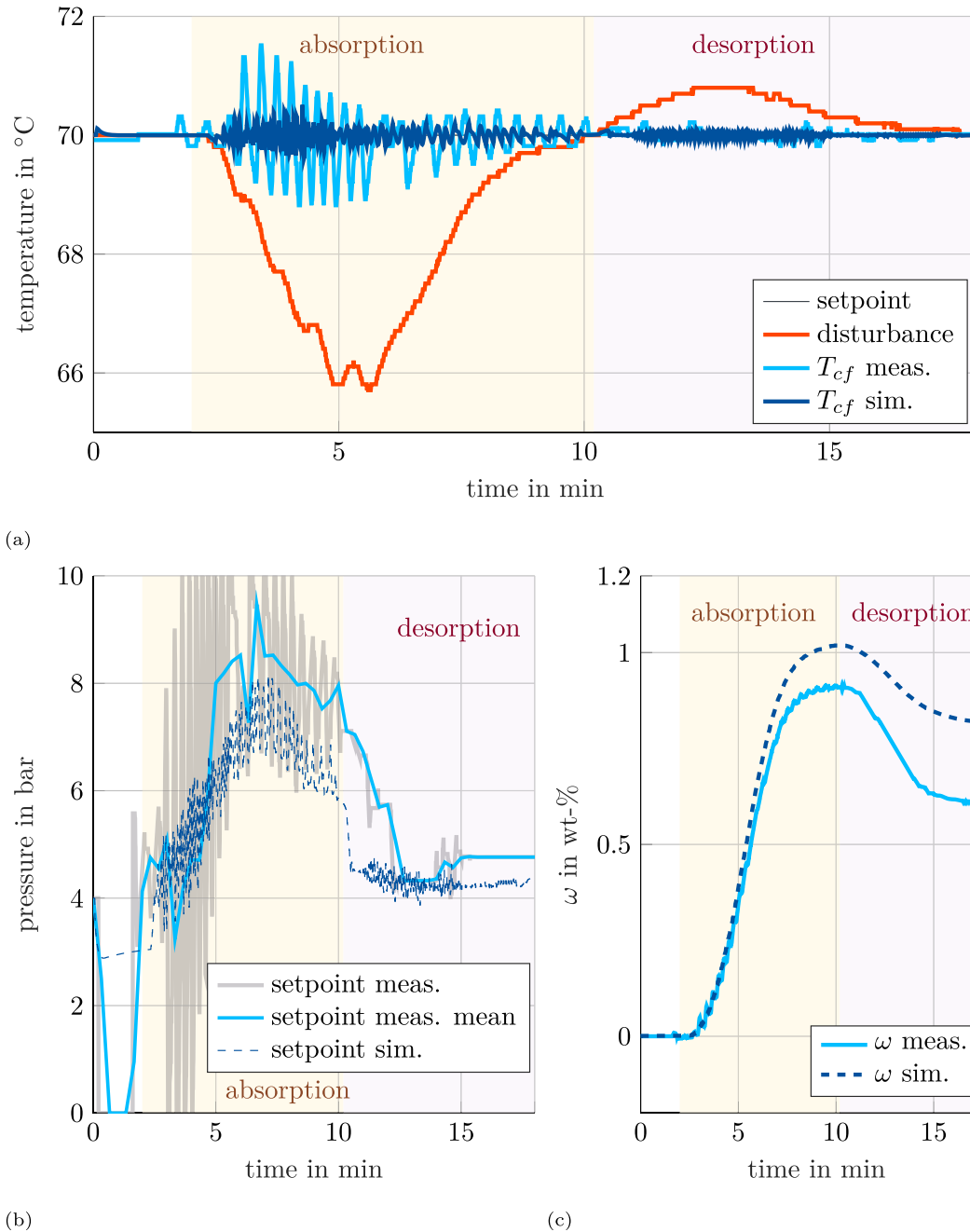


Fig. 9 – Results of the simulation and experiment with the proposed control structure. (a) Temperature of the cooling fluid at the inlet and outlet of the reactor during simulation and measurement. (b) Pressure p_{set} from the simulation and the measurement data. (c) Hydrogen charging of the control loop simulation and experiment.

especially during the absorption phase, are due to non-modeled effects in the reaction rate and the pressure regulators.

Finally, Fig. 9(c) shows the hydrogen charging for simulation and experiment. During the first half of the absorption phase the measurement and simulation data agree well. Then, from $t = 8$ min to $t = 10$ min, the two curves start to deviate. For the subsequent desorption phase, this deviation increases. The reason for this lies in differences between the approximated material data to the real behaviour of the metal hydride.

Concluding this experimental validation, it can be summarized that despite the deviations between model and

experiment, especially regarding the reaction rate, the model was able to capture the crucial features of the MH behaviour. Thus, it was possible to design a controller which was able to realize an experimental proof of concept for the active thermal buffer module based on MHs.

3.5. General discussion of the active thermal buffer module

In order to realize an active thermal buffer element, next to the demonstration of the stabilization of the temperature, it is

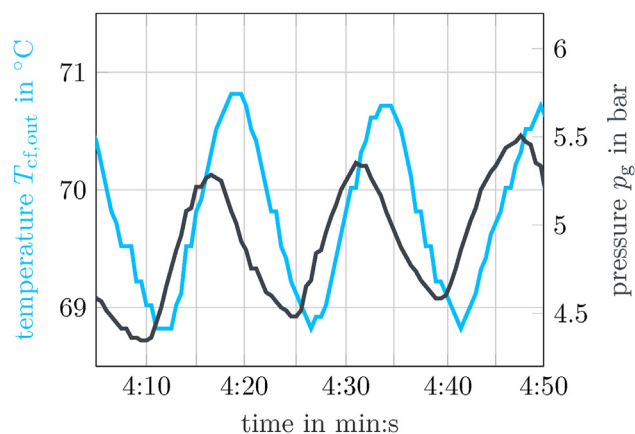


Fig. 10 – Time sequence to compare the pressure p_g in the reactor and the temperature T_{cf} at the outlet of the reactor.

of special interest to evaluate the achieved response time. Thus, to demonstrate how fast a variation in the set pressure results in a response in the outlet temperature of the fluid. Therefore, the time sequences of the measured pressure p_g and temperature T_{cf} are shown in Fig. 10 in detail for an arbitrarily selected time period. It is obvious that a variation in pressure leads to a temperature change within a maximum delay of 5 s, which is fast for a thermal energy storage system. Concluding, the experiments have proven that the dynamic of the MH even inside the reactor are very fast and thus, the gas pressure is a suitable control variable for the design of an active thermal buffer based on MHs.

Next to the generic response time of the system, it is also important to determine the achieved thermal peak power as well as the achieved amount of thermal energy that has been provided by the system. In the present case, the thermal peak power can be calculated to 724.8 W for absorption and 134.8 W for desorption or 755 W/kg_{MH} and 140 W/kg_{MH}, respectively. It has to be considered, that the desorption peak power was limited by the framework of the test bench. Looking at the thermal energy calculated with (2) evaluated at every discrete time point and summed up, the reactor could compensate the thermal disturbance with 146.96 kJ during absorption and –31.62 kJ during desorption. Due to the smaller temperature amplitude and the same time length as in the absorption disturbance, the released thermal energy during desorption is lesser.

When designing such an active thermal buffer, the sizing of the module can be a limitation for its wide applicability. As the system is based on thermal capacity, the compensation of e.g., increased thermal loads when going uphill is rather not realistic as it would require very high masses. In this case, the additional weight of an increased heat exchanger removing the occurring thermal energy immediately, will be more useful. However, due to the possible response times and the achieved thermal power, the system can be included as small module to compensate for temperature peaks in the range of seconds, which would in turn afford high efforts in control of the tardy overall heat transfer fluid circuit.

4. Conclusion

In the present study, an experimental setup has been realized to demonstrate the temperature control in a heat transfer fluid circuit using an active thermal buffer reactor based on MHs, which requires neither hydrogen nor electricity. For this, a simple PI controller was designed based on validated 0D-model simulations. Using the gas pressure as actuating variable, the PI controller was able to transfer fluctuating inlet temperatures with –3.9 K and +1.2 K to a constant outlet temperature. A temperature disturbance with the thermal energy of ~147 kJ and ~–32 kJ has been compensated at thermal powers of +755 W/kg_{MH} and 140 W/kg_{MH} for absorption and desorption, respectively.

The features of such a system are:

- Active control of the outlet temperature for fluctuating inlet temperatures by actuation of the gas pressure.
- High peak powers 755 W/kg_{MH} for heating are measured. Due to reaction enthalpies of more or less same size during absorption and desorption, similar peaks for cooling can be expected.
- Response times below 5 s of the system following the gas pressure due to immediate reaction and good heat transfer properties

Future work will focus on an improved model quality by improved design of the measurement concept and on a advanced controller, e.g., with a feed forward part. For application, the achieved specific thermal power will assist in designing and sizing suitable systems and thereby it will help to evaluate the possibility of a cooler size reduction in an existing fuel cell cooling system.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Measurement equipment overview

An overview of the used measurement equipment is given in Table A.3.

Table A.3 – Overview of the measurement equipment at the test bench.

parameter	sensor name	manufacturer's designation	range	deviation
temperature T	$T_{c,siev}$	type K, class 1	–40 ... 1000	±1.5 K
	$T_{c,cf}$ & $T_{c,cf,in}$ calibrated to each other	type K, class 1	10 ... 95 °C	±0.05 K
pressure p	P_1	Keller PA-21-y	1 ... 160 bar	±0.8 bar
	P_2	integrated in PR ₁ (Brooks SLA5810)	0.120 bar	–
	P_3	Keller PA-21-y	1 ... 30 bar	±0.15 bar
	P_4	Keller PA-21-y	1 ... 60 bar	±0.3 bar
mass flow $\dot{m}$	$\dot{M}_{H_2O}$	Promass 83F	0 ... 2000 kg h ^{–1}	±0.1%
	$\dot{M}_{H_2}$	Bronkhorst F–201CV–5F6 PGD-33-Y	0 ... 5 L _N min ^{–1}	

Appendix B. Calculation of the hydrogen charging value

In the following, the calculation of the hydrogen charging value is described in more detail.

During absorption ω is calculated with the help of the Sieverts volume with the defined volume $V_{siev} = 2.25$ l and the measured values T_{siev} and p_{siev} . With those parameters and the approach

$$n_{siev}(t) = \frac{p_{siev}(t)V_{siev}}{RT_{siev}(T)Z(p_{siev}(t), T_{siev}(t))}, \quad (B.1)$$

with the fugacity coefficient Z and the real gas coefficient R , the actual amount of hydrogen in the volume n_{siev} can be calculated. The value of the coefficient Z is calculated according to the approach in Ref. [30]. With the amount of hydrogen in the Sieverts volume n_{siev} , the amount of hydrogen in the void volume n_{tot} , the absorbed amount of hydrogen n_{abs} can be calculated as

$$n_{abs}(t) = n_{siev}(t_0) - n_{siev}(t) - n_{tot}(t). \quad (B.2)$$

Furthermore, the hydrogen charging state is calculated with

$$\omega = \frac{n_{H_2}}{M_{H_2}m_{MH}} \cdot 100\%, \quad (B.3)$$

where n_{H_2} represents the amount of hydrogen, M_{H_2} the molar mass of hydrogen and m_{MH} the mass of MH in the reactor.

During desorption the hydrogen charging can be measured with PR₂ for hydrogen flow rates below 5 Nl/min and recalculated with the ideal gas law to a hydrogen loading rate. For higher volume flow rates the hydrogen charging is estimated by the relation between pressure and the hydrogen charging as

$$\frac{\omega(t) - \omega(t-1)}{\Delta\omega_{ges}} = \frac{p(t) - p(t-1)}{\Delta p_{ges}}, \quad (B.4)$$

where $\Delta\omega_{ges}$ and Δp_{ges} represent the changes between the initial and end values during a desorption reaction. Hence, an equality of the dynamics in the loading and pressure is assumed. The initial values for the calculation are taken from the previous reaction. The final charging is approximated via the approximated PCI's.

Appendix C. Calculation of the hydrogen mass stream into and out of the reactor

The mass flow rate $\dot{m}_p$ into the reactor in Eq. (12) is not known. As an approach the polytropic and reversible adiabatic outflow equation of an ideal gas

$$\dot{m}_p = A_g \psi \sqrt{\frac{2p_g}{RT_{amb}}} \quad (C.1)$$

is set up with the outflow function ψ approximated as

$$\psi \approx \psi_{fit} = -0.4854 \tanh\left(5.2\left(\frac{p_p}{p_g} - 1\right)\right). \quad (C.2)$$

In the equations, A_g represents the surface of the outlet cross section of the reactor, p_g the gas pressure in the reactor, p_p the gas pressure in the pipes. The detailed procedure of the derivation can be found in Ref. [31].

Appendix D. Calculation of the equilibrium pressure in the 0D - Model

In the following section, the calculation of the equilibrium pressure in Chapter 2.2 is discussed more in detail. Starting with the calculation of the required reaction enthalpies $\Delta H_{R,abs}$, $\Delta H_{R,des}$ and the entropies ΔS_{abs} , ΔS_{des} . Those are based on the evaluation of the PCI data in Ref. [24] for the hydrogen loadings of $\omega_{rel} = [10\%, 50\%, 90\%]$. To calculate the enthalpy and entropy values for arbitrary loadings, linear regression

$$\Delta\mathcal{X} = \varphi_1\omega_{rel} + \varphi_2 \quad (D.1)$$

is used. Here, $\Delta\mathcal{X}$ is a placeholder for the enthalpy and entropy values, φ_1 and φ_2 determine the coefficients, and ω_{rel} is the present charging state. The values of the coefficients are listed in Table D.4.

Table D.4 – Coefficients of the linear regression in Eq. D.1 for the reaction enthalpy and entropy values.

	reaction	φ_1	φ_2
ΔH_R	Absorption	–13.375	$3.17 \cdot 10^4$
	Desorption	–2.75	$3.395 \cdot 10^4$
ΔS_R	Absorption	–0.005	105.18
	Desorption	0.027	108.551

Finally, the calculated loading-dependent enthalpy and entropy values are used to approximate the equilibrium pressure p_{eq} . Therefore, the Van't Hoff equation

$$\ln\left(\frac{p_{eq}}{p_0}\right) = \begin{cases} \frac{\Delta H_{R,abs}(\omega)}{RT_{MH}} + \frac{\Delta S_{R,abs}(\omega)}{R} & \text{Absorption} \\ \frac{\Delta H_{R,des}(\omega)}{RT_{MH}} + \frac{\Delta S_{R,des}(\omega)}{R} & \text{Desorption} \end{cases} \quad (D.2)$$

distinguishes between absorption (abs) and desorption (des). In the equation, T_{MH} represents the temperature of the MH, p_0 the standardized pressure, R the ideal gas constant and, as described before, ΔH_R and ΔS_R the reaction enthalpy and entropy. Eq. (D.2) describes the so called plateau region of the typical progression of the standardized and logarithmically scaled pressure over the hydrogen load in the MH, called pressure concentration isotherm (PCI). The plateau is limited by steep climbs of the pressure for the nearly empty or fully charged MH. In the modeling, the steeper increase of the equilibrium pressure at higher hydrogen loadings during absorption and the steep decrease of the pressure for lower hydrogen loadings have to be included. Therefore, Eq. (D.2) is adapted during absorption with

$$I_{add} = a(3)\coth(a(1)(\omega - a(2)\omega_{max})^2) - 1, \quad (D.3)$$

and the coefficient values

$$a = [12.8499 \quad 1.0197 \quad 0.1608] \quad (D.4)$$

where ω_{max} is the maximum hydrogen charging value. For desorption the steep decreases are simplified and ω is limited to values greater than zero. With those supplements Eq. (13) is derived.

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